

Let $V := H_0^1(\Omega)$ and let $a(\cdot, \cdot) : V \times V \rightarrow \mathbb{R}$ be bounded and coercive with constants $C, c > 0$ as stated, and let $F \in V' := H^{-1}(\Omega)$. Consider first the continuous variational problem: find $y \in V$ such that $a(y, v) = F(v)$ for all $v \in V$. Since a is bilinear, bounded, and coercive on V , the Lax–Milgram Lemma implies existence of a unique solution $y \in V$.

Next fix a finite-dimensional subspace $V_h^0 \subset V$ and consider the discrete problem: find $y_h \in V_h^0$ such that $a(y_h, v_h) = F(v_h)$ for all $v_h \in V_h^0$. Because V_h^0 is a subspace of V , the same boundedness and coercivity inequalities hold when u, v are restricted to V_h^0 : indeed, for $u, v \in V_h^0$ we have $|a(u, v)| \leq C\|u\|_{1,\Omega}\|v\|_{1,\Omega}$ and $a(u, u) \geq c\|u\|_{1,\Omega}^2$. Also F is a bounded linear functional on V , so its restriction to V_h^0 is a bounded linear functional in $(V_h^0)'$. Therefore the Lax–Milgram Lemma applied to the Hilbert space V_h^0 with the inherited norm $\|\cdot\|_{1,\Omega}$ gives existence and uniqueness of $y_h \in V_h^0$.

For the error estimate, let $e := y - y_h \in V$ and let $v_h \in V_h^0$ be arbitrary. Define $w_h := y_h - v_h \in V_h^0$. Subtracting the Galerkin equations gives, for every $w_h \in V_h^0$,

$$a(e, w_h) = a(y - y_h, w_h) = F(w_h) - F(w_h) = 0.$$

In particular, with $w_h = y_h - v_h$ we obtain $a(e, y_h - v_h) = 0$, and by linearity in the second argument,

$$a(e, y - v_h) = a(e, (y - y_h) + (y_h - v_h)) = a(e, e) + a(e, y_h - v_h) = a(e, e).$$

Thus $a(e, e) = a(e, y - v_h)$.

Using coercivity and boundedness we estimate

$$c\|e\|_{1,\Omega}^2 \leq a(e, e) = a(e, y - v_h) \leq C\|e\|_{1,\Omega}\|y - v_h\|_{1,\Omega}.$$

If $e = 0$, the desired inequality holds trivially. Otherwise divide by $\|e\|_{1,\Omega}$ to get

$$\|y - y_h\|_{1,\Omega} = \|e\|_{1,\Omega} \leq (C/c)\|y - v_h\|_{1,\Omega}.$$

Since $v_h \in V_h^0$ was arbitrary, taking the infimum over all $v_h \in V_h^0$ yields

$$\|y - y_h\|_{1,\Omega} \leq (C/c) \inf_{v_h \in V_h^0} \|y - v_h\|_{1,\Omega}.$$

This proves Cea’s Lemma along with the claimed error bound and the uniqueness of y and y_h .